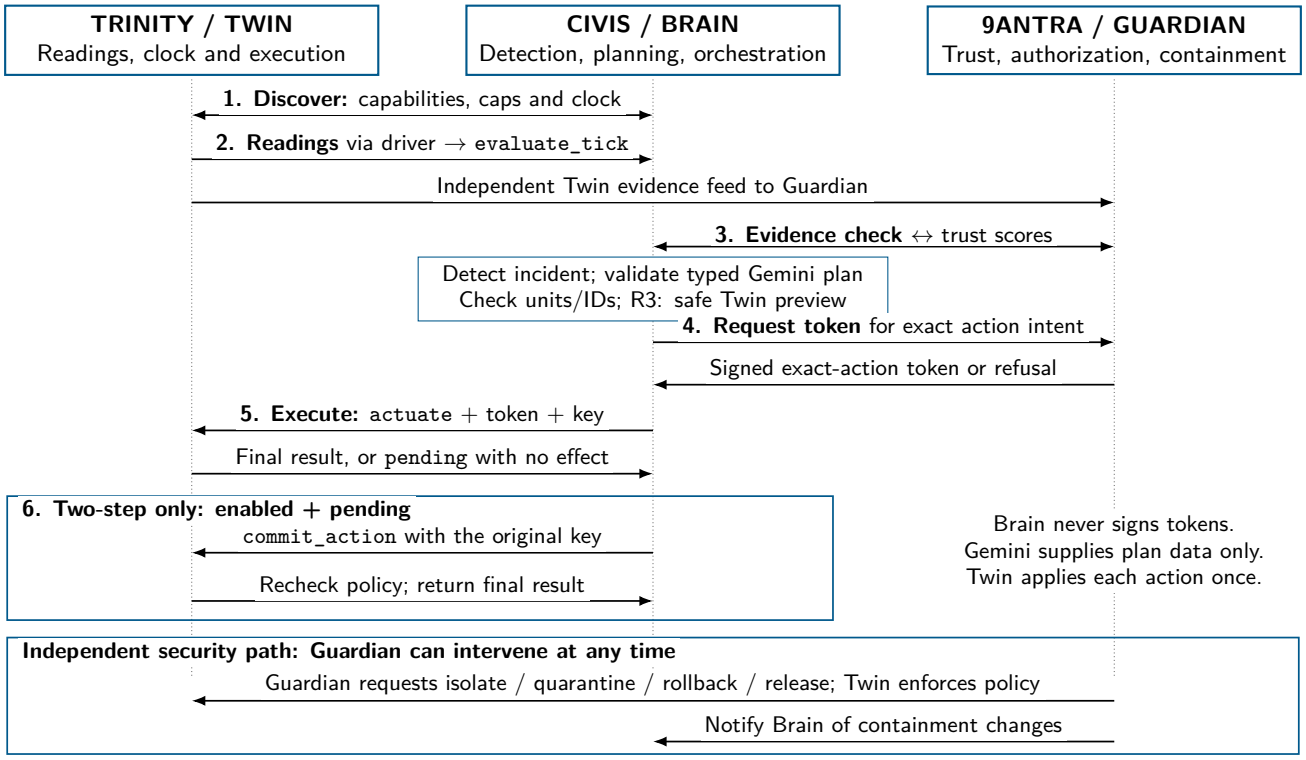
Three-layer architecture and workflow

CIVIS design for partner confirmation / 16 October call



Connections: MCP Streamable HTTP (2026-07-28) for tools; SSE for feeds. Separate caller Bearer keys authenticate peers. Guardian holds the signing private key. Scores never replace tokens. **Actuation fields:** `action`, `targets`, `params`, `token`, `idempotency_key`. Twin checks token, expiry, containment and caps before execution.

Clocks and timeout policy

Operation	Bound / clock	Behaviour
Peer MCP	3 real seconds/call	Monotonic timeout, limited by remaining phase budget. Released mock uses 5 s.
Gemini planning	60 real seconds	One request, no automatic retry; before authorization. Timeout blocks action.
Evidence recovery	10 real seconds total	Neighbours; same sensor after 1 simulated tick (wait max 3 real s); fresh Guardian verdict. At most three steps.
Pending lifetime	10 real seconds	Twin monotonic clock from acceptance. Simulation pause/speed does not extend it; token may expire sooner.
Commit confirmation	3 real seconds/call	Call immediately. On lost reply only: one same-key retry after 1 real s, within deadline. At most 7 s overall.
Token lifetime	60 simulated seconds	Agreed lifetime; Twin clock is authoritative. No local wall-clock conversion.

Recovery: finish evidence recovery before authorization. Refusal blocks the action; a lost reply leaves its outcome unknown. Never use a new key or extend a deadline to bypass uncertainty.

Traceability and delivery status

Each layer writes `run-logs/<run_id>.<layer>.jsonl`; collect and merge all three per run. Link reading IDs, token ID and idempotency key. Never log full tokens, secrets or hidden true values.

Integration gates: released mock confirmation runs on a later evaluation; prompt confirmation, this timeout/retry policy and Guardian's pushed-score feed require joint integration. Continuous-clock planning currently blocks when Twin advances. Keep two-step off until joint tests pass. Guardian's T5 reaction target is tested separately from pending expiry.